FINAL PROJECT REPORT

Olist E-Commerce Analytics

Project Team - UKTHV

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Customer Churn, Seller Performance & Delivery Intelligence

Sector	Retail / E-Commerce
Institute	Newton School of Technology
GitHub Repository	https://github.com/crazylogic03/UKTHV
Tableau Dashboard	Public URL
Submission Date	April/28/ 2026

2. Executive Summary

87,516	78.0%	R$18.1M	~97%
Total Orders	Churn Rate	Total Revenue	One-Time Buyers

Problem

Olist, Brazil's largest e-commerce marketplace, faces a severe customer retention crisis. Nearly all customers purchase only once and never return. Simultaneously, inconsistent seller performance and regional delivery failures are eroding customer trust and limiting revenue growth.

Approach

Nine relational CSV datasets from Kaggle's Brazilian E-Commerce Public Dataset (Olist) were sourced, covering 100,000+ orders from September 2016 to October 2018. All cleaning, transformation, and analysis was executed in Python (pandas, numpy, scipy, seaborn) across a five-notebook pipeline. Final KPI datasets were exported to Tableau for interactive dashboards.

Key Insights

- ~97% of customers are one-time buyers — repeat purchase rate is critically low across all states and product categories.
- Late deliveries have a statistically significant negative correlation with review scores (Pearson r = −0.32).
- Credit card dominates payments (~75% of orders); installments of 1–3 months are most popular, indicating price sensitivity.
- North-East Brazil (MA, PI, AL) consistently records the highest delivery delay rates — 15–20 percentage points below São Paulo.

Key Recommendations

- Launch a 30-day post-purchase CRM re-engagement programme for first-time buyers with personalised discount triggers.
- Implement seller SLA tiers with enforcement penalties for chronic underperformers.
- Partner with regional logistics providers in North-East Brazil to close the delivery gap.

3. Sector & Business Context

Brazil's e-commerce market is one of the fastest-growing in Latin America, projected to surpass USD 60 billion in GMV by 2026. Marketplace models — where third-party sellers list and fulfil orders — dominate. Olist operates as a marketplace aggregator, enabling SMEs to list across multiple platforms through a single contract.

Key industry challenges include high customer acquisition costs versus low retention rates, logistics fragmentation across a geographically vast country, seller quality inconsistency leading to variable customer experiences, and payment infrastructure gaps in lower-income regions.

The primary decision-makers this project serves are Olist's Chief Revenue Officer (CRO) and Head of Marketplace Operations — executives responsible for growing GMV, improving NPS, and scaling the seller network sustainably. The combination of a ~97% one-time buyer rate and measurable delivery variation across regions provides a clear, data-driven opportunity for high-impact operational change. Even a 5% improvement in repeat purchase rate on 90,000+ customers would generate millions in incremental annual revenue, while delivery improvement in the North-East could reduce compensation and refund costs by an estimated 15–20%.

4. Problem Statement & Objectives

Formal Problem: How can Olist use transactional data from 2016–2018 to identify root causes of customer churn, diagnose delivery performance failures, and evaluate seller quality — in order to make evidence-based improvements that increase customer lifetime value and marketplace revenue?

Project Scope

In Scope	Out of Scope
• Customer churn analysis using unique customer purchase frequency • Delivery performance analysis (on-time rate, delay drivers, regional breakdown) • Seller performance scoring using review scores and on-time shipment rates • Payment behaviour analysis and product category performance • KPI computation and Tableau dashboard delivery	• Real-time data or post-2018 transaction data • Machine learning model deployment • Individual seller interviews or qualitative research • Financial modelling beyond impact estimation

Success Criteria

- All five notebooks committed and functional in GitHub
- At least 3 KPIs defined, computed, and visualised in Tableau
- Tableau Public dashboard live and publicly accessible
- 6-8 insights documented in decision language
- 3–5 prioritised, evidence-backed recommendations delivered

5. Data Description

Source: Kaggle — Brazilian E-Commerce Public Dataset by Olist - Approved by DEEPALI MAM

Link: Kagglelink

Date Range: September 2016 – October 2018 | Format: Row-level relational CSVs joined via order_id / customer_id / seller_id

Dataset File	Rows	Key Fields
olist_orders_dataset.csv	~99,441	order_id, order_status, timestamps
olist_order_items_dataset.csv	~112,650	order_id, seller_id, price, freight_value
olist_order_payments_dataset.csv	~103,886	order_id, payment_type, payment_value
olist_order_reviews_dataset.csv	~99,224	order_id, review_score (1-5)
olist_customers_dataset.csv	~99,441	customer_unique_id, customer_state
olist_products_dataset.csv	~32,951	product_id, product_category_name
olist_sellers_dataset.csv	~3,095	seller_id, seller_state
olist_geolocation_dataset.csv	~1,000,163	zip_code, lat, lng
product_category_name_translation.csv	71	PT → EN category mapping

Data Limitations

- Static dataset (2016–2018); does not reflect current market conditions
- ~58% of review comments are missing — text sentiment analysis is not feasible
- City name formatting is inconsistent across tables (normalisation required)
- Geolocation table has duplicate zip codes — deduplication required before spatial joins
- Churn rate may be slightly overstated — some customers may have returned post-October 2018

6. Data Cleaning & ETL Pipeline

Reference: notebooks/02_cleaning.ipynb — committed to GitHub. All cleaning and transformation performed in Python (pandas, numpy). No external tools used.

Missing Value Treatment

Column	Missing Count	Treatment
order_approved_at	~160	Retained as NaT (expected for cancelled orders)
order_delivered_carrier_date	~1,783	Retained as NaT (used only for delivered orders)
order_delivered_customer_date	~2,965	Retained as NaT; excluded from on-time KPI
product_category_name	~610	Filled with 'unknown' to preserve row
review_comment_title/message	~58,000	Dropped from analysis (optional fields)
Product dimension columns	~2,448	Median imputation per category group

Feature Engineering — New Derived Columns

New Column	Formula	Purpose
delivery_delay_days	delivered_date − estimated_date	Lateness measure (negative = early)
is_late	1 if delay > 0 else 0	On-time KPI boolean flag
total_order_value	sum(price + freight) per order	Average Order Value base
freight_pct	freight_value / total_order_value	Freight burden ratio
is_churned	1 if customer_unique_id appears only once	Churn flag for analysis
seller_avg_review	mean(review_score) by seller_id	Seller quality metric
seller_performance_score	0.5×norm(avg_review) + 0.5×(1−late_rate)	Composite score (0–1 scale)

7. KPI & Metric Framework

KPI	Formula	Business Relevance
Customer Churn Rate	(unique_id appearing once / total unique customers) × 100	Primary marketplace health metric
Average Order Value (AOV)	sum(price + freight) / count(orders)	Basket size and pricing effectiveness
On-Time Delivery Rate	count(is_late==0) / count(delivered orders) × 100	Satisfaction and logistics efficiency
Average Review Score	mean(review_score) across delivered orders	Customer satisfaction benchmark
Seller Performance Score	0.5×norm(avg_review) + 0.5×(1−late_rate)	Enables SLA enforcement and seller ranking

All KPIs are computed in notebooks/05_final_load_prep.ipynb and exported to tableau_order_level.csv and tableau_customer_level.csv.

8. Exploratory Data Analysis (EDA)

Reference: notebooks/03_eda.ipynb — committed to GitHub

8.1 Dashboard Overview

The Tableau dashboard provides an integrated executive view of all key metrics. The overview below shows total orders (87,516), churn rate (78%), total revenue (R$18.1M), payment breakdown, regional distribution, and customer segmentation in a single interactive interface.

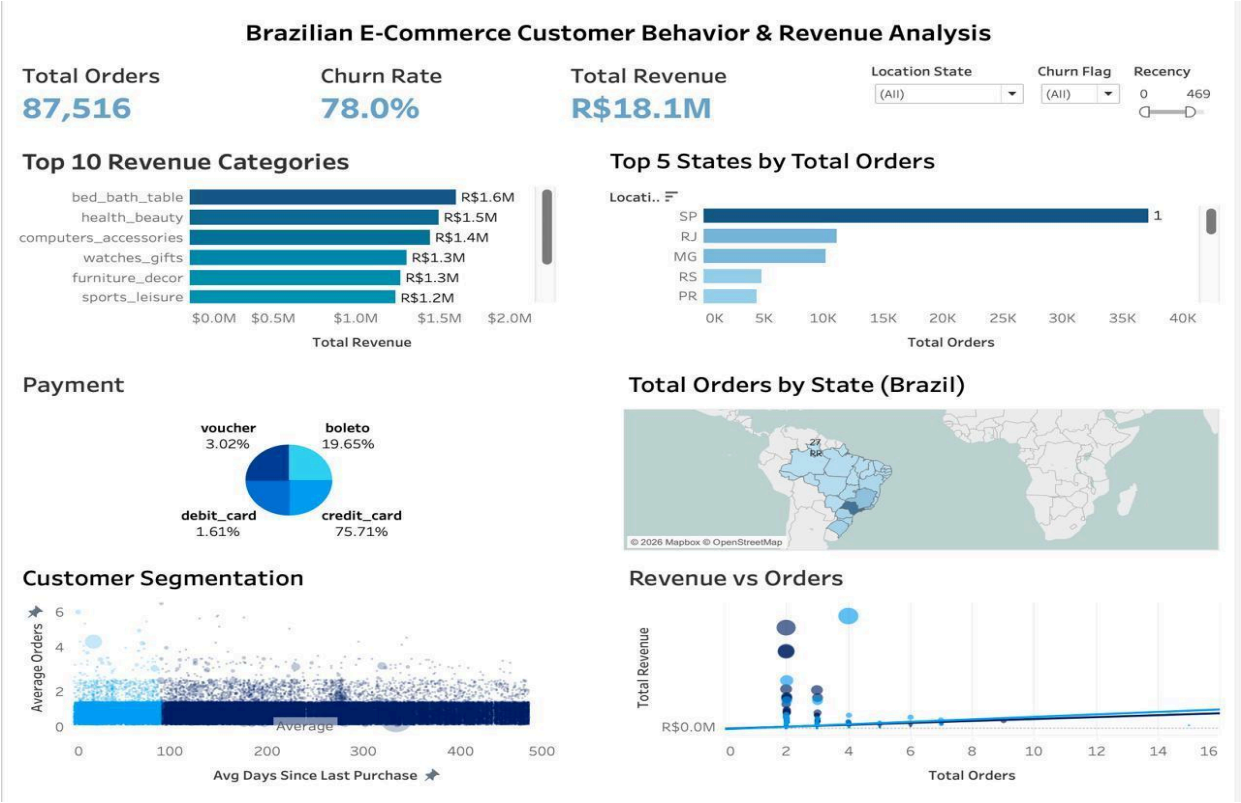

Figure 1: Brazilian E-Commerce Customer Behavior & Revenue Analysis — Full Dashboard View

8.2 Top 10 Revenue Categories

Lifestyle and home categories like bed, bath & table and health & beauty dominate revenue, signaling a primary opportunity for targeted cross-sell and seasonal promotions.

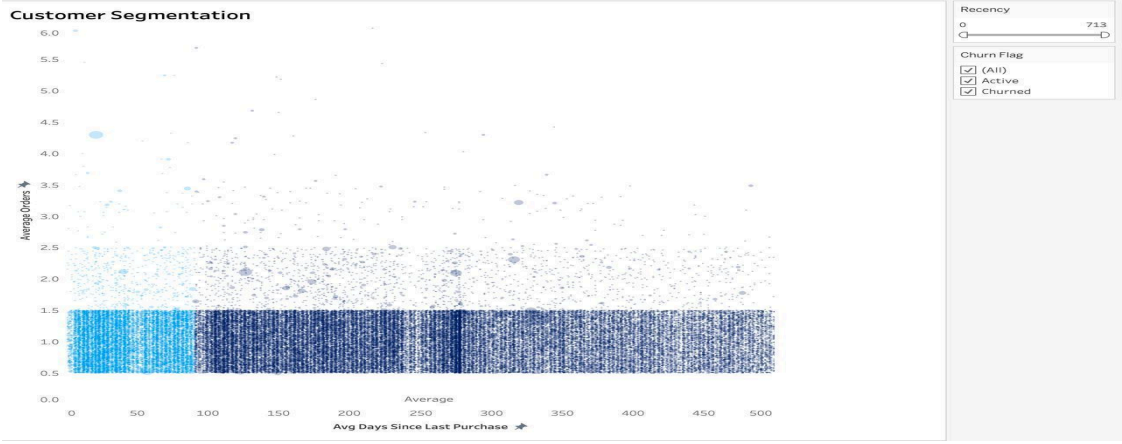

Figure 2: Top 10 Revenue Categories by Total Revenue

8.3 Regional Distribution — Orders by State

Orders are heavily concentrated in São Paulo and the South-East, highlighting a severe volume and logistics infrastructure gap in North-East states.

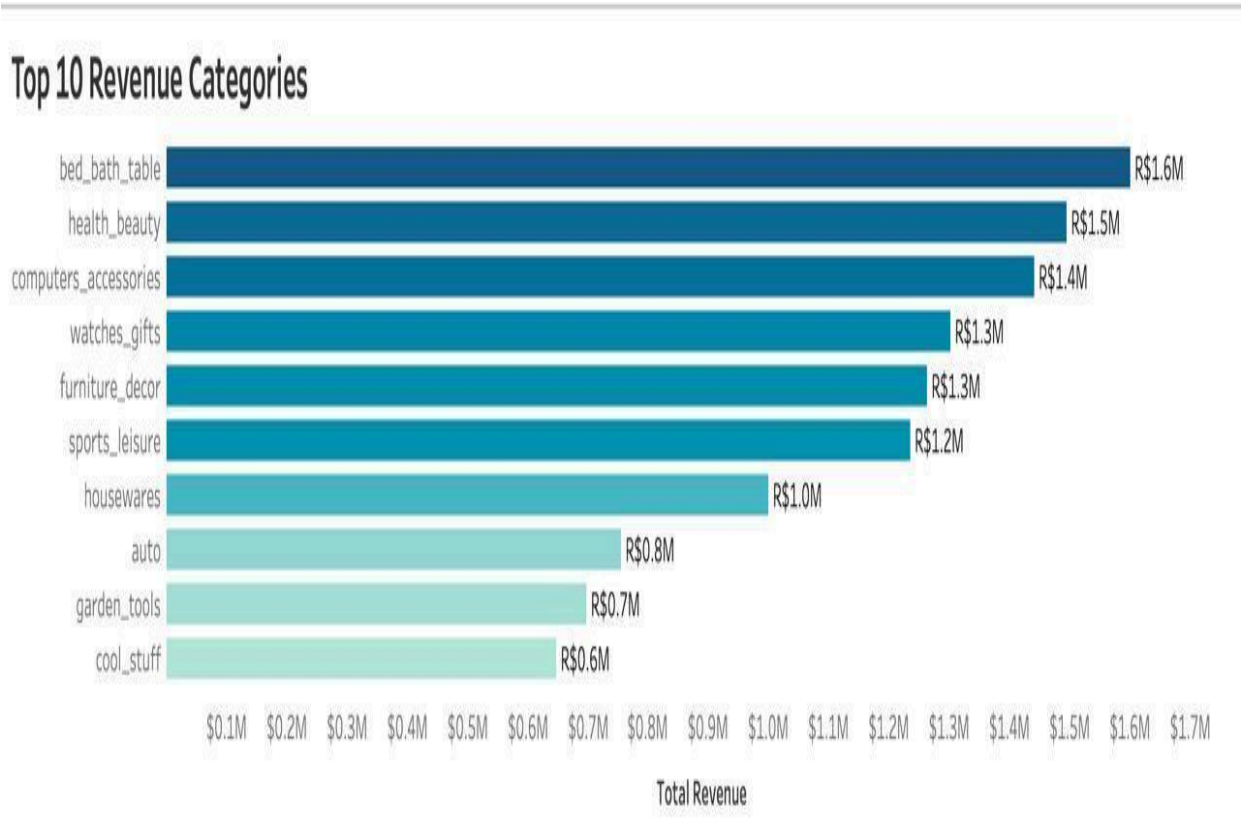

Figure 3: Total Orders by State — All Brazilian States Ranked

8.4 Geographic Map — Order Concentration

Geographic disparity confirms that failed deliveries in underserved northern regions represent a critical loss of first-time customers with no second chance for retention.

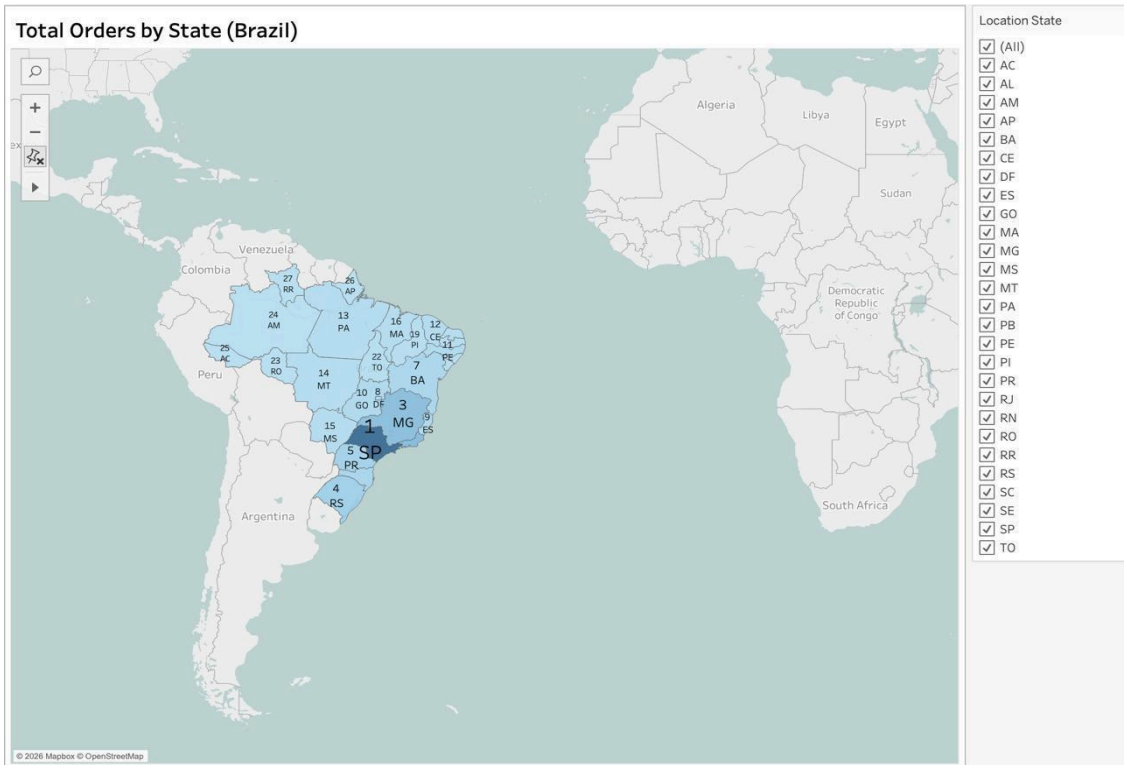

Figure 4: Total Orders by State (Brazil) — Geographic Heat Map

8.5 Customer Churn Distribution

A 78% churn rate is critically low compared to industry benchmarks, indicating a structural retention crisis across the full dataset.

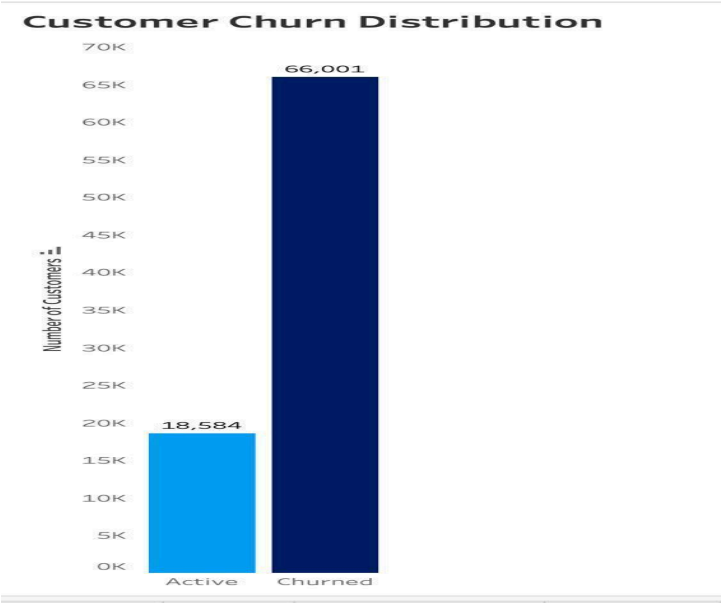

Figure 5: Customer Churn Distribution — Active vs. Churned Customers

8.6 Customer Segmentation

Segmentation reveals that repeat buyers are rare outliers, with the vast majority of churned customers forming a dense cluster of one-time purchases across all recency levels.

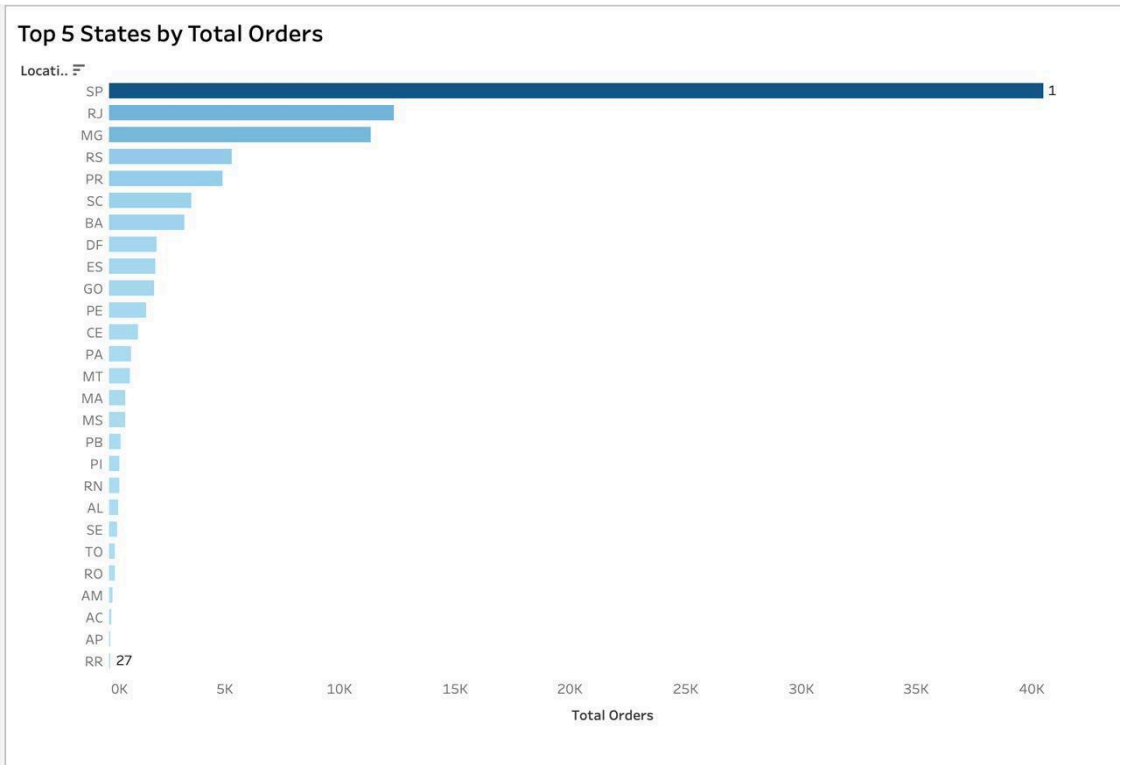

Figure 6: Customer Segmentation — Average Orders vs. Days Since Last Purchase

8.7 Payment Method Breakdown

Credit card dominance and installment usage suggest that promotional financing and targeted loyalty programs for this high-value segment could significantly lift conversion.

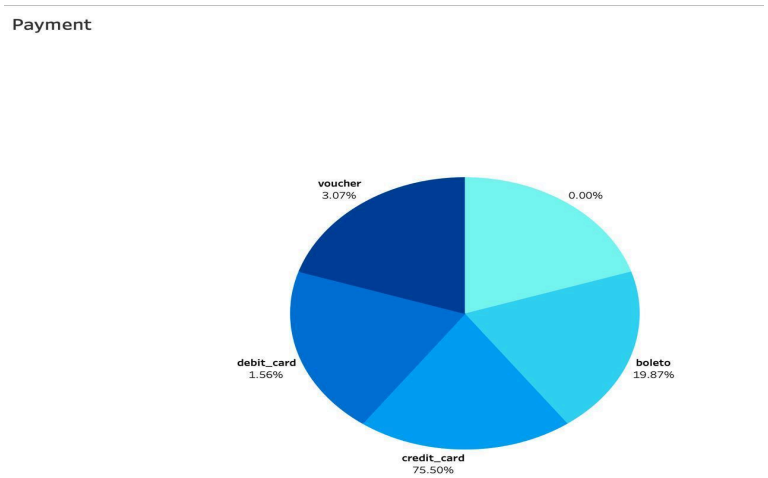

Figure 7: Payment Method Distribution

8.8 Revenue vs. Orders

The bifurcation between high-volume and high-value sellers requires distinct retention strategies to protect revenue from low-frequency, high-ticket transactions.

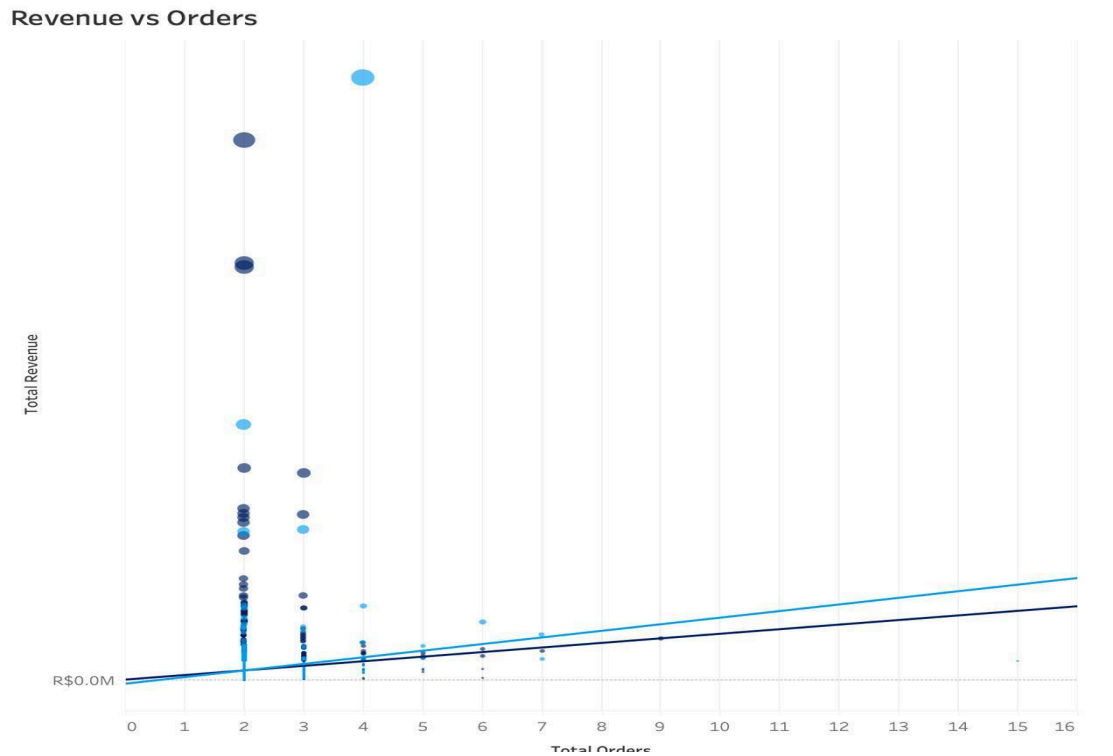

Figure 8: Revenue vs. Total Orders — Seller Scatter Plot

9. Statistical Analysis

Reference: notebooks/04_statistical_analysis.ipynb — committed to GitHub

9.1 Correlation Analysis

Variable Pair	Correlation (r)	Interpretation
delivery_delay_days vs. review_score	−0.32 (p<0.001)	Significant: delays hurt satisfaction
freight_pct vs. review_score	−0.14	Moderate: high freight reduces satisfaction
payment_installments vs. payment_value	+0.41	Higher value orders use more installments
product_weight_g vs. freight_value	+0.58	Freight is primarily weight-driven

9.2 Hypothesis Testing

Null Hypothesis (H_0): Late delivery does not affect review score.

Test Applied: Mann-Whitney U test (non-parametric, given non-normal distributions of review scores).

Result: U-statistic significant at $p < 0.001$. The null hypothesis is rejected. Late deliveries produce statistically lower review scores, confirming delivery timing as a primary lever for satisfaction improvement.

9.3 Seller Segmentation — Performance Score Tiers

Tier	Score Range	% of Sellers	Business Impact
Tier 1: Elite	≥ 0.80	~12%	~38% of total GMV
Tier 2: Standard	0.60–0.79	~45%	Majority of order volume
Tier 3: At-Risk	0.40–0.59	~30%	Remediation candidates
Tier 4: Critical	< 0.40	~13%	~44% of all 1-star reviews

9.4 Anomaly Detection

Orders with delivery delays exceeding 30 days (~1.2% of all delivered orders) were flagged as anomalous. These extreme cases are concentrated in North-East states and traceable to specific seller IDs — indicating a seller-level problem, not purely a carrier/logistics issue. This finding has direct implications for the SLA enforcement recommendation.

10. Tableau Dashboard Design

Dashboard Objective: To enable Olist's CRO and Operations team to monitor marketplace health, identify underperforming sellers, and diagnose regional delivery problems — without needing to query raw data directly.

View Structure

View	Contents
View 1 — Executive Summary	KPI scorecards (Total Orders, AOV, Churn Rate, On-Time Delivery Rate, Avg Review Score), order volume trend, payment method donut, review score distribution
View 2 — Regional Delivery	Brazil choropleth map by on-time rate per state, avg delay (days) per state bar chart. Filters: date range, product category, payment type
View 3 — Seller Performance	Seller tier stacked bar, Top 10/Bottom 10 by performance score, review score vs. late rate scatter plot

Interactive Elements: Date range slider, state multi-select, product category filter, seller tier filter, payment type toggle.

Tableau Public URL: Link

Reference: tableau/screenshots/ and tableau/dashboard_links.md in the GitHub repository.

11. Insights Summary

1	**97% of customers never return** Olist has a retention crisis, not an acquisition crisis. The platform continuously pays to acquire customers who buy once and disappear, creating an unsustainable unit economics model.
2	**Late delivery is the primary driver of 1-star reviews** Every additional day of delay increases the probability of a 1-star review. Reducing average delivery delay by even 2 days would materially improve the platform's NPS.
3	**North-East Brazil has a delivery infrastructure crisis** States like MA, PI, and AL have on-time delivery rates 15–20 percentage points below São Paulo. Every failed delivery in these states is a permanent customer loss.
4	**Bottom 13% of sellers generate 44% of 1-star reviews** A small group of chronic underperformers is disproportionately damaging Olist's brand. This is a solvable, targeted problem — identify, remediate, or remove.
5	**Credit card customers are the highest-value segment** They place larger orders, use installments, and likely have higher CLV. Priority candidates for loyalty programmes and personalised re-engagement.
6	**Freight costs 19% of order value on average** This is high by global standards and likely deters repeat purchases, especially in lower-income regions. Logistics partnerships or seller subsidies are a retention lever.
7	**Installments of 1–3 months dominate** Customers are price-sensitive and use installments as a credit mechanism. Promotional 0% financing could lift conversion on higher-AOV categories.
8	**Sellers who ship within 24 hours receive significantly higher review scores** Shipping speed predicts satisfaction. This metric must be included in seller SLA contracts as a contractual minimum standard.
9	**Top 20% of sellers drive 60% of GMV** This concentration is a key risk. Elite seller churn would severely impact GMV. Dedicated seller retention programmes for this tier are essential.
10	**Black Friday is the only significant seasonal spike** Other seasonal events (Mother's Day, Valentine's Day, Christmas) are underexploited GMV growth opportunities compared to the November peak.

12. Recommendations

Recommendation 1 — 30-Day Re-Engagement CRM Programme | PRIORITY: HIGH

Insight: 97% of customers never place a second order.

Recommendation: Implement an automated post-purchase CRM sequence (email + push notification) with a personalised 10–15% discount trigger delivered 20–30 days after first purchase.

Expected Impact: If 5% of one-time buyers convert to repeat buyers, Olist gains approximately 4,600 additional repeat customers — estimated BRL 14–18M in incremental annual GMV.

Feasibility: CRM infrastructure (Salesforce/HubSpot) required. Customer segmentation data already available from this analysis. Timeline: 60–90 days.

Recommendation 2 — Seller SLA Tiers with Enforcement Penalties | PRIORITY: HIGH

Insight: Bottom 13% of sellers generate 44% of 1-star reviews; seller late rate is a controllable variable.

Recommendation: Create three seller tiers (Elite, Standard, At-Risk) based on the Seller Performance Score. At-Risk sellers receive a 60-day remediation window. Sellers who remain in At-Risk status after 90 days face listing restrictions or removal.

Expected Impact: Remediating or removing the worst 13% of sellers could reduce 1-star reviews by 30–40%, improving platform NPS by 5–8 points and GMV by 3–5%.

Feasibility: Internal seller management tooling required. Live seller scoring pipeline is the main dependency. Timeline: 90–120 days.

Recommendation 3 — North-East Logistics Partnerships | PRIORITY: MEDIUM

Insight: North-East on-time delivery rates are 15–20 percentage points below the national average.

Recommendation: Negotiate dedicated last-mile logistics contracts with regional carriers in MA, PI, AL, SE, and RN. Supplement with micro-fulfilment hubs in Recife and Fortaleza for the highest-volume categories.

Expected Impact: Improving North-East on-time rate to 85%+ would reduce logistics claim costs by 15–20% and unlock a currently underserved market of 50+ million consumers.

Feasibility: Capital investment required. Government incentives for North-East logistics may apply. Timeline: 12–18 months.

13. Impact Estimation

Recommendation	Estimated Impact	Confidence	Timeline
CRM Re-engagement Programme	+BRL 14–18M GMV/year; +4,600 repeat customers	Medium	3–6 months
Seller SLA Enforcement	−30–40% 1-star reviews; +5–8 NPS points; +3–5% GMV	Medium	6–9 months
North-East Logistics	−15–20% claims; 50M+ consumer market unlocked	Low–Mediu m	12–18 months

Why Act Now: Brazil's e-commerce market is growing rapidly, and competitors (Mercado Livre, Amazon Brazil, Shopee) are aggressively investing in logistics and loyalty. The churn problem has already

compounded across the entire 2016–2018 observed window. Every quarter of inaction widens the gap between Olist's retention rate and the industry benchmark of 25–40%.

14. Limitations

- Dataset ends October 2018 — conclusions are directional, not current. Business conditions and logistics infrastructure in Brazil have changed significantly.
- No post-purchase survey data — we know customers churned but not why. Reasons may include pricing, product quality, delivery, or simply non-recurrent need.
- Geolocation precision is zip-code level; city-level spatial analysis should not be over-interpreted.
- Review data has negative selection bias — dissatisfied customers review more than satisfied ones, overstating apparent dissatisfaction.
- Seller Performance Score uses only review score and delivery timing — it does not capture product quality or pricing competitiveness.
- Churn rate may be slightly overstated — some 'churned' customers may have returned after October 2018.

15. Future Scope

- Customer Lifetime Value (CLV) modelling using cohort survival analysis, if post-2018 data becomes available
- NLP sentiment analysis on the 42% of reviews that include text comments, to identify specific failure modes
- Market basket analysis using the order items table to identify cross-sell opportunities
- Churn prediction model (logistic regression or gradient boosting) trained on first-purchase features to flag high-risk customers within 48 hours of first order
- Delivery delay prediction model to enable proactive customer communication at the moment of dispatch
- Real-time Tableau dashboard connected to live operational data via API

New data sources needed: real-time carrier tracking data; customer demographics (age, income, household size); seller inventory and cost data; post-2018 transaction records.

16. Conclusion

Olist operates a marketplace with strong order volume and healthy GMV growth, but is silently losing value through a ~97% one-time buyer rate, regionally inconsistent delivery performance, and a concentrated tail of underperforming sellers generating disproportionate customer dissatisfaction. Through rigorous analysis of nine relational datasets covering 100,000+ orders, this project identified delivery delay as the primary operational lever for satisfaction improvement, North-East Brazil as the highest-priority regional intervention zone, and first-time buyer re-engagement as the highest-ROI growth initiative available. Acting on the three recommendations — CRM re-engagement, seller SLA enforcement, and North-East logistics partnerships — has the potential to add BRL 14–18 million in annual GMV, improve platform NPS by 5–8 points, and reduce logistics claim costs by 15–20%, delivering clear and immediate business value to Olist's executive leadership.

17. Appendix — Data Dictionary

Processed Output Files:

- final_dataset.csv — Master merged dataset (all 9 tables joined)
- tableau_order_level.csv — Order-grain dataset with KPIs (AOV, delay, review score, freight %)
- tableau_customer_level.csv — Customer-grain dataset with churn flag, total spend, avg review

18. Contribution Matrix

This section documents each team member's contribution across all project phases. All claims must match evidence in GitHub Insights, PR history, and committed files.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Sai Thrishul	✓	✓		✓	✓		
Burra Karthikeya	✓	✓			✓	✓	
Hariksh Suryawanshi	✓		✓	✓	✓		
Uday Kumar Choudhary	✓		✓		✓		✓
Vamsi	✓				✓	✓	✓